Lecture 5

Van Kampen, surfaces and braids

Objectives. *We compute loop classes in a space assembled from open pieces. Van Kampen's theorem records how loops in the overlap become relations. For a surface presented by a polygon, its boundary gives the relation. For identical particles, the space consists of unordered configurations of distinct points. Loops in this space describe exchanges. Planar exchanges retain more information than their endpoint permutations. Under a scalar statistical-transport assumption, these exchanges allow arbitrary unit phases.*

By the end of this lecture you will be able to:

- Apply van Kampen with path-connected open sets and intersection.
- Read the fundamental-group presentation of a polygonal surface.
- Relate planar exchanges to braids and three-dimensional exchanges to permutations.
- Derive the allowed one-dimensional exchange phases in each case.

Unit 5A Surfaces

5.1 Free groups and relations

Definition 5.1 (Wedge of circles). The *wedge* $S^1 \vee S^1$ identifies one point of each of two circles to a common point x_0.

We call the loops round the two circles a and b. A loop can traverse these circles in an ordered sequence. Traversing a circle and immediately retracing it gives a removable inverse pair. The following figure records the order of traversals, including the four traversals of the word $aba^{-1}b^{-1}$, called the *commutator*.

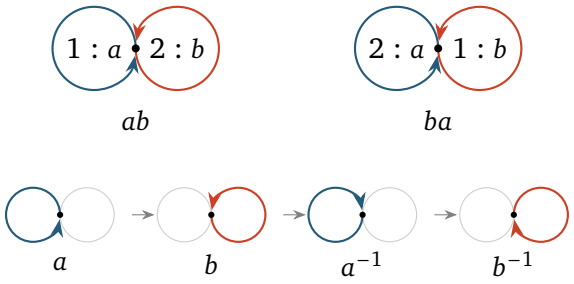


Figure 5.1: The numerals in ab and ba give the order of traversal. The commutator traverses each circle once in each direction from the common point.

Definition 5.2 (Word and reduced word)**.** A *word* in letters $a_1, \dots, a_k$ and their formal inverses is a finite string of these symbols. A *reduced word* has no adjacent inverse pairs.

Delete adjacent inverse pairs until none remain. The resulting reduced word is unique. One algorithm reads from left to right, keeping a stack: cancel the last stored letter when the incoming letter is its inverse, and append it otherwise. Inserting an adjacent inverse pair leaves the final stack unchanged, even when the first inserted letter cancels a stored letter. The second then restores it. Thus reduction is independent of the order of deletions. A reduced word passes through the stack unchanged. Hence every complete sequence of deletions ends at the same stack output.

Definition 5.3 (Free group)**.** The *free group* F_k consists of reduced words. Multiply by concatenating and reducing. The empty word is the identity. Reverse a word and invert each letter to get its inverse.

Reduction is unchanged by inserting or deleting inverse pairs in any context, so reducing after either stage of a triple concatenation gives the same result. This proves associativity. For example,

$$ab^{-1}ba^{-1}b = b, \qquad aba^{-1}b^{-1} \neq 1 \quad \text{in } F_2.$$

The second word is already reduced. A choice of images for the generators in any group extends uniquely to a homomorphism, by substituting in words. Inverse cancellations do not change the substituted group element.

Recall from Lecture 4 that a subgroup contains the identity and is closed under products and inverses. The subgroup generated by a set consists of finite products of its elements and their inverses.

Definition 5.4 (Kernel)**.** The *kernel* of a homomorphism $f\colon G \to H$ is the subgroup

$$\ker f = \{g \in G : f(g) = 1_H\}.$$

The homomorphism laws make this set closed under products and inverses. They also give $f(gng^{-1}) = 1_H$ for $g \in G$ and $n \in \ker f$.

Definition 5.5 (Normal subgroup)**.** A subgroup N of G is *normal* if $gNg^{-1} = N$ for every $g \in G$.

Thus every kernel is normal. Applying the preceding calculation with g^{-1} gives equality of the conjugate subgroup with the kernel.

Definition 5.6 (Coset and quotient group)**.** For a normal subgroup N of G, the *coset* of $g \in G$ is

$$gN = \{gn : n \in N\}.$$

The *quotient group* G/N is the set of cosets, with product

$$(gN)(hN) = ghN.$$

Two cosets agree exactly when $g^{-1}h \in N$. If $h = gn$ with $n \in N$, then $hN = gnN = gN$. Conversely, equality of the cosets gives $h \in hN = gN$, so $h = gn$ for some $n \in N$. To check independence of representatives, write $g' = gn$ and $h' = hm$ with $n, m \in N$. Then

$$g'h' = gnhm = gh(h^{-1}nh)m \in ghN.$$

Normality puts $h^{-1}nh$ in N. Hence the product is well defined. Associativity follows from

associativity in G. The identity coset is N, and the inverse of gN is $g^{-1}N$.

A surjective homomorphism $f\colon G \to H$ with kernel N identifies G/N with H by $gN \mapsto f(g)$. Indeed, $f(g) = f(h)$ holds exactly when $g^{-1}h \in N$. This proves that the induced map is both well defined and injective. It is surjective because f is surjective, and it preserves products. For any homomorphism the same argument identifies $G/\ker f$ with its image.

Definition 5.7 (Normal closure)**.** The *normal closure* of specified words is the smallest normal subgroup containing them, equivalently the subgroup generated by all their conjugates and inverses.

Imposing a relation $r = 1$ also imposes $uru^{-1} = 1$ for every word u. This explains why the whole normal closure is imposed.

Definition 5.8 (Presentation)**.** A *presentation* is written

$$\langle a_1, \dots, a_k \mid r_1, \dots, r_s \rangle = F_k / \langle\langle r_1, \dots, r_s \rangle\rangle.$$

It imposes $r_j = 1$ in every word context, not merely deletion of one printed copy of a relator. For example, $\langle a \mid a^m \rangle \cong \mathbb{Z}_m$ for $m \geq 1$, where $\mathbb{Z}_m$ is the group of integers modulo m. For a two-generator example, impose $aba^{-1}b^{-1} = 1$. Multiplying on the right by ba gives $ab = ba$. Thus every word becomes $a^m b^n$ for $m, n \in \mathbb{Z}$. The exponent-sum map to $(m, n) \in \mathbb{Z}^2$ kills the relator and is inverse to $(m, n) \mapsto a^m b^n$. Hence

$$\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

In the free group the commutator remains a nonempty reduced word.

Definition 5.9 (Free product)**.** The *free product* $G * H$ joins two groups without additional relations: words alternate nonidentity elements of the two groups, combining adjacent elements from the same group.

In particular, $\mathbb{Z} * \mathbb{Z} = F_2$.

5.2 Van Kampen's theorem

Van Kampen uses loops from both open pieces as generators and identifies the two images of every overlap loop. Let $X = U \cup V$, with $x_0 \in U \cap V$. The inclusions induce maps from the fundamental groups of these sets. Write i_{U*} and i_{V*} for the maps from the intersection to the two pieces.

Theorem 5.10 (Van Kampen · *cited; surjectivity sketched*)**.** *Suppose U and V are open and path-connected, and $U \cap V$ is path-connected. The natural homomorphism*

$$\pi_1(U, x_0) * \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0)$$

is surjective. Its kernel is the normal closure of $i_{U*}(w) i_{V*}(w)^{-1}$ for $w \in \pi_1(U \cap V, x_0)$.

This is Hatcher, Theorem 1.20. It identifies the two images of each overlap loop. The quotient by the stated kernel is therefore the group of X, by the quotient construction in §5.1. We sketch the surjectivity construction. Subdivide a loop into pieces γ_j in U or V, using the Lebesgue-number lemma. Merge successive pieces in the same set. At a change of set the joining point is in $U \cap V$. Choose a path a_j in the intersection from x_0 to each joining

point, with constant paths at the start and end. The based loops $a_{j-1} \cdot \gamma_j \cdot \bar{a}_j$ each lie in one piece. Their product represents the original loop, since the inserted reverse paths cancel. The kernel statement is cited, not proved by this construction.

The connected-intersection hypothesis matters. On S^1, take the north point $N = (0, 1)$ and south point $S = (0, -1)$. Cover the circle by their complements. Both sets are open intervals and have trivial fundamental group. Their intersection has two components. The circle group is $\mathbb{Z}$, so the conclusion of the theorem cannot hold without the hypothesis.

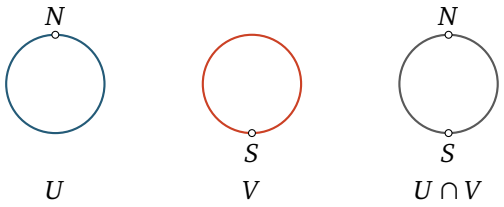


Figure 5.2: Removing N or S gives an open arc. Removing both leaves two components, so this circle cover does not satisfy van Kampen's hypotheses.

For the wedge of two circles, remove a non-base point y_b of the second circle to get U, and a non-base point y_a of the first to get V. These are open in the wedge. The first retracts onto the first circle, the second onto the second circle. Collapse the remaining open arcs linearly to x_0; these homotopies fix their retained circles. The formulas agree at x_0, so finite closed pasting proves continuity. The intersection is four open arms joined at x_0 and contracts there. Hence the cited theorem gives

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

The loops a and b run round the two circles. Inductively the same open-neighbourhood construction gives F_r for a finite wedge of r circles. The figure marks the points removed to construct the two open sets.

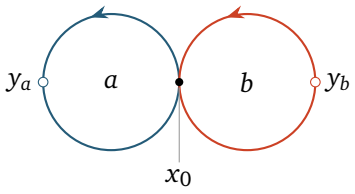


Figure 5.3: Removing the marked point from the opposite circle leaves an open neighbourhood that retracts onto the chosen circle. Both circles share x_0.

For a sphere S^n with $n \geq 2$, remove its north pole to obtain U and its south pole to obtain V. Both pieces are homeomorphic to $\mathbb{R}^n$, so their groups are trivial. Their overlap is path-connected: it is homeomorphic to $S^{n-1} \times \mathbb{R}$. Van Kampen gives a trivial fundamental group for S^n. This recovers Lecture 4's sphere result from the cited theorem.

For $\mathbb{R}\mathrm{P}^2$, use Lecture 2's disc with antipodal boundary points identified. Let U be its quotient minus the centre and let V be a small open centre disc. The set U retracts radially onto the boundary quotient, a circle with generator a. The overlap is a punctured disc. Its counter-clockwise generator runs round the full disc boundary, so runs twice round the boundary quotient. Its image in U is a^2, and its image in V is trivial. Thus van Kampen gives

$$\pi_1(\mathbb{R}\mathrm{P}^2) \cong \langle a \mid a^2 \rangle \cong \mathbb{Z}_2.$$

This agrees with the antipodal-cover calculation in Lecture 4.

5.3 The torus square

We compute the group of the square with opposite sides identified in the same direction. The quotient is the torus T^2 of Lecture 2. All four corners give one vertex x_0, and the two edge loops are a and b. Choose the centre o in the face. Let $U = T^2 \setminus \{o\}$ and let V be a small open disc about o inside the face. These are path-connected open sets covering the torus. The set U retracts radially onto the edge wedge, and V is contractible. Their overlap is the punctured disc $V \setminus \{o\}$, whose group is $\mathbb{Z}$.

Choose a base point y in this overlap and a connector from the vertex x_0 to y in U. We orient the overlap generator counter-clockwise in the displayed square, rather than as an inner boundary of a punctured surface. Following the connector, the overlap loop and the reverse connector gives a loop based at x_0 in U. The figure shows its radial retraction to the square boundary.

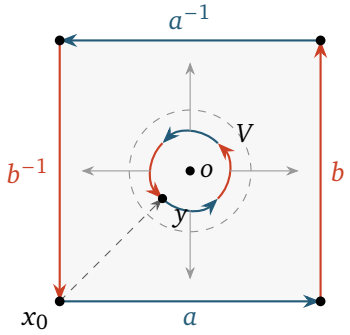


Figure 5.4: Edge arrows show the counter-clockwise boundary traversal $aba^{-1}b^{-1}$ of the retracted overlap loop, with opposite edges identified in the same direction. The dashed connector joins the vertex to its base point y.

The overlap generator maps to $aba^{-1}b^{-1}$ in the edge wedge and to the identity in the disc. A change of base point along the connector identifies the wedge group with $\pi_1(U, y)$. Van Kampen imposes exactly the normal closure of this relator. Therefore

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

The last isomorphism is the two-generator quotient computed in §5.1. This agrees with the coordinate-winding calculation of Lecture 3.

5.4 Polygonal surfaces

Definition 5.11 (Surface). A *surface* is a Hausdorff space with a countable basis in which every point has a neighbourhood homeomorphic to an open disc in $\mathbb{R}^2$. A compact surface with these local models is called *closed*.

Definition 5.12 (Genus surface). For $g \geq 1$, let Σ_g be the standard closed surface obtained from a $4g$-gon by pairing equally labelled edges in the boundary word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

The integer g is its *genus*.

The torus has one handle. Joining g tori along boundary circles of removed discs gives the surface with g handles. Its identification with the polygon model is cited from Hatcher, §1.2. The following figure shows the genus-two model and its polygon. The numbers $0, \dots, 7$ name the octagon corners. In the inset we rescale the corner sectors to show their cyclic order around the identified vertex. On the surface, the dashed paths join a common point x_0 to each displayed curve. Following a connector, its curve and the reverse connector gives the based generator with the corresponding polygon label.

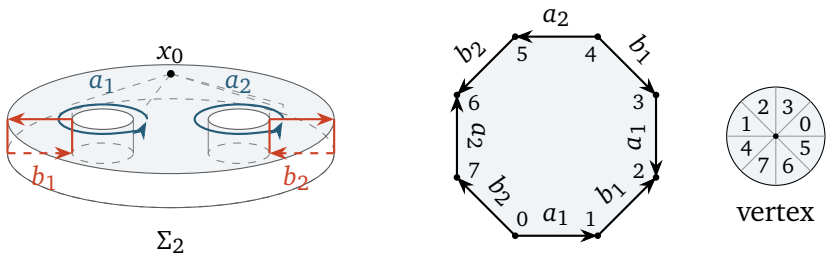


Figure 5.5: A surface with two handles and its octagon model. The paired edges give four loops, and the corner sectors join to one disc.

The paired arrows specify the same direction along each identified edge. Edge interiors join two half discs, and the cyclic corner identifications join to a disc. All vertices become one vertex, and the edges become a wedge of $2g$ circles. This is the polygonal model used in Hatcher, §1.2.

Proposition 5.13 (Surface groups · *proved given van Kampen*)**.** *For $g \geq 1$, the fundamental group is*

$$\pi_1(\Sigma_g) \cong \left\langle a_1, b_1, \dots, a_g, b_g \;\middle|\; \prod_{j=1}^{g} [a_j, b_j] \right\rangle, \qquad [a, b] = aba^{-1}b^{-1}.$$

Proof. Choose a point o in the face interior. Let $U = \Sigma_g \setminus \{o\}$ and let V be a small open disc about o inside that face. The punctured polygon retracts radially to its boundary, fixing it. This deformation respects the edge identifications and retracts U onto the wedge of $2g$ circles. The set V is contractible. The overlap is a punctured disc, with fundamental group $\mathbb{Z}$. Its positive generator follows the face boundary once after radial retraction. Choose a path from its base point to the polygon vertex to read the word in the stated generators. The result is $[a_1, b_1] \cdots [a_g, b_g]$. Van Kampen kills exactly its normal closure, giving the presentation. A different connecting path conjugates the relator and leaves its normal closure unchanged. ∎

For $g = 1$ this is the torus presentation computed in §5.3.

Definition 5.14 (Permutation group)**.** A permutation of $\{1, \dots, n\}$ is a bijection of this set to itself. These form the group S_n, with right-to-left composition. In cycle notation, (12) exchanges 1 and 2, while (123) sends 1 to 2, 2 to 3 and 3 to 1.

Self-check. Prove that the group of $S^1 \vee S^1$ is nonabelian by finding a surjective homomorphism onto S_3.

Answer. Send a to (12) and b to (13). The universal property of F_2 gives a homomorphism. Its image contains $(23) = (12)(13)(12)$ and the two three-cycles, as well as the identity.

Thus it contains all six permutations. Since $(12)(13) \neq (13)(12)$, the image is nonabelian, and F_2 is nonabelian. In particular, its commutator $aba^{-1}b^{-1}$ cannot be removed before a face imposes that relation. Its image is $(132)^2 = (123) \neq 1$, which verifies nontriviality directly.

Self-check. Suppose $X = U \cup V$ satisfies van Kampen's hypotheses, with $\pi_1(U) = \langle a \rangle \cong \mathbb{Z}$, $\pi_1(V) = \langle b \rangle \cong \mathbb{Z}$ and $\pi_1(U \cap V) = \langle w \rangle \cong \mathbb{Z}$. The inclusion maps send w to a^3 and b^2 respectively. Give a presentation of $\pi_1(X)$. Determine whether assigning $a \mapsto e^{2\pi \mathrm{i}/7}$ and $b \mapsto e^{3\pi \mathrm{i}/7}$ defines a homomorphism to the group of unit complex numbers.

Answer. The two images of w must agree, so

$$\pi_1(X) \cong \langle a, b \mid a^3 b^{-2} \rangle.$$

A homomorphism from the free group descends to this quotient exactly when it kills the relator. For the proposed images, its value on the relator is

$$(e^{2\pi \mathrm{i}/7})^3 (e^{3\pi \mathrm{i}/7})^{-2} = e^{(6-6)\pi \mathrm{i}/7} = 1.$$

Thus the assignment does define a homomorphism.

Unit 5B Particles and braids

5.5 Configuration spaces and endpoint permutations

We model n identical particles by their positions in $\mathbb{R}^d$. We restrict to histories with no coincident positions. An ordered list keeps labels, whereas its unordered set forgets them. Fix integers $n \geq 2$ and $d \geq 2$.

Definition 5.15 (Configuration spaces)**.** The *ordered configuration space* and *unordered configuration space* are

$$\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d) = \{(x_1, \dots, x_n) \in (\mathbb{R}^d)^n : x_i \neq x_j \text{ for } i \neq j\},$$
$$\mathrm{Conf}_n(\mathbb{R}^d) = \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d)/S_n.$$

We keep the dimension in the notation to distinguish the plane from space. In the planar case we also write $C_n = \mathrm{Conf}_n(\mathbb{R}^2)$. The permutation group S_n acts by relabelling the coordinates. Its action is $(s \cdot x)_i = x_{s^{-1}(i)}$, which forgets labels in the quotient. The projection is a covering: take disjoint small balls around the n points. Their product and its $n!$ permuted copies are disjoint sheets over the same quotient neighbourhood. The quotient map is open because the inverse image of the image of an open set is the finite union of its permuted copies.

For two particles in either dimension, write their positions as $c + u$ and $c - u$, where $c \in \mathbb{R}^d$ and $u \in \mathbb{R}^d \setminus \{0\}$. The inverse formulas are $c = (x_1 + x_2)/2$ and $u = (x_1 - x_2)/2$. The centre is unrestricted. Collision occurs exactly when $u = 0$. Forgetting labels identifies u with $-u$. The centre, radius and line through u therefore give a homeomorphism

$$\mathrm{Conf}_2(\mathbb{R}^d) \cong \mathbb{R}^d \times (0, \infty) \times \mathbb{R}\mathrm{P}^{d-1}.$$

The inverse sends $(c, r, [v])$, with $|v| = 1$, to $\{c + rv, c - rv\}$. This is independent of the representative v. Local sections of the antipodal cover make the inverse continuous.

In space, the last factor is $\mathbb{R}\mathrm{P}^2$. Its fundamental group is $\mathbb{Z}_2$ by Lecture 4, given the sphere's simple connectedness. One exchange lifts to a half great circle from v to $-v$ on S^2. Two exchanges lift to the full great circle. The following figure compares this loop with the planar one.

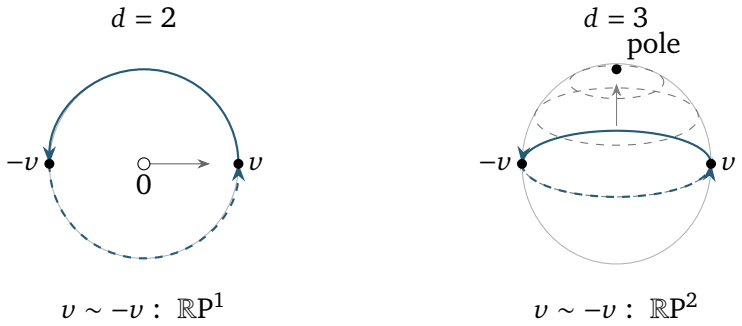


Figure 5.6: One exchange joins opposite directions, identified in projective space. The full circle contracts on S^2 through latitudes, whereas the planar double exchange has winding two in the squared direction and does not contract.

For an explicit free contraction of the lifted spatial double exchange, set

$$H(t,s) = \big(\cos(\pi s/2)\cos(2\pi t),\ \cos(\pi s/2)\sin(2\pi t),\ \sin(\pi s/2)\big), \qquad t,s \in [0,1].$$

Each latitude lies on the unit sphere. At $s = 0$ it is the great circle, and at $s = 1$ it is the north pole. The base point moves during this contraction. Attaching a path along the moving base point and its reverse, as in Lecture 3, converts it into a based contraction. Composing with the antipodal projection and keeping the centre and radius fixed contracts the double exchange in $\mathrm{Conf}_2(\mathbb{R}^3)$.

For two planar particles, write the positions as $c+u$ and $c-u$ in $\mathbb{C}$, where $u \neq 0$. Forgetting labels identifies u with $-u$. The centre c, radius $|u|$ and squared direction $(u/|u|)^2$ give

$$\mathrm{Conf}_2(\mathbb{R}^2) \cong \mathbb{R}^2 \times (0,\infty) \times S^1.$$

The inverse chooses either square root of the direction; the two choices give the same unordered pair. Locally a square-root branch is continuous, so the inverse is continuous. Contracting the first two factors gives $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) \cong \mathbb{Z}$. The counter-clockwise exchange $u(t) = e^{\pi i t}$ makes the squared direction wind once. Two exchanges have winding two, even though their endpoint permutation is the identity.

A loop downstairs lifts to n labelled particle trajectories. Label their initial positions $1, \dots, n$.

Definition 5.16 (Outgoing permutation)**.** The *outgoing permutation* $p_\gamma(i)$ is the final position of the particle starting at i.

With the first-path-first convention,

$$p_{\gamma\cdot\delta} = p_\delta \circ p_\gamma, \qquad \rho([\gamma]) = p_\gamma^{-1}, \qquad \rho([\gamma][\delta]) = \rho([\gamma])\rho([\delta]).$$

Paths run first to last, while permutations compose right to left. Taking the inverse reverses the order again, so ρ is a homomorphism. Homotopy lifting makes it well-defined on loop classes. For the noncommuting test $p_\gamma = (12)$ and $p_\delta = (13)$, it gives

$$p_{\gamma\cdot\delta} = (123), \qquad \rho([\gamma][\delta]) = (132) = (12)(13).$$

In the plane, fix an unordered base configuration x_0.

Definition 5.17 (Geometric braid and braid group)**.** A *geometric braid* is n collision-free trajectories $(z_j(t), t)$ with that fixed set of positions at both $t = 0$ and $t = 1$. Time increases along every strand. Deformations keep the endpoint positions fixed and preserve the collision-free condition. Write B_n for their classes, with multiplication by running the first braid and then the second.

Proposition 5.18 (Exchange classes · *sketched; proved in* $\mathbb{R}^3$)**.** *In three dimensions,* $\pi_1(\mathrm{Conf}_n(\mathbb{R}^3)) \cong S_n$. *In the plane, the fundamental group is the braid group* B_n.

For $\mathbb{R}^3$, we prove that the ordered space is simply connected. We approximate a loop by a polygon, then contract it to a suitable apex. The dimension bound allows a single apex avoiding every pairwise collision. Its collision sets are linear subspaces $L_{ik} = \{x_i = x_k\}$, for $1 \le i < k \le n$, of $\mathbb{R}^{3n}$, each of dimension $3n - 3$. A finite union of proper linear subspaces misses a point. Indeed, for each subspace choose a nonzero linear functional vanishing on it. On the curve $(1, t, \dots, t^{3n-1})$ each functional becomes a nonzero polynomial, with finitely many roots. Choose t outside those finite root sets.

First, a loop γ has positive distance ε from the closed union of collision sets, by compactness. Uniform continuity gives a subdivision $0 = t_0 < \cdots < t_m = 1$ with each image piece of diameter less than $\varepsilon/4$. Join successive vertices $v_j = \gamma(t_j)$ by straight segments to form η. At matching parameters, the interpolation between η and γ stays within $\varepsilon/2$ of $\gamma(t)$. This is a based homotopy avoiding collisions.

Next, choose an apex z_* outside all subspaces $L_{ik} + \mathrm{span}\{v_j, v_{j+1}\}$. Each has dimension at most $3n - 1$, so the finite-union argument applies. With two spatial coordinates, that sum can fill the ambient space, so the apex argument does not apply. The triangle with vertices v_j, v_{j+1}, z_* misses L_{ik}. Otherwise a collision point with nonzero apex coefficient would express z_* as an element of that forbidden subspace. A collision with zero apex coefficient would lie on the already collision-free polygon edge. Thus the map $h\colon [0,1]^2 \to \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^3)$ given by

$$h(t, s) = (1 - s)\eta(t) + sz_*$$

contracts the loop freely. Lecture 3's moving-base-point construction turns this into a based contraction by attaching the path of the moving base point and its reverse. Apply the same apex choice to two configurations instead of polygon vertices to join them by two straight segments. Hence the ordered space is path-connected and simply connected. The relative-coordinate triangle below displays the collision-free cone for one edge and contrasts it with a planar triangle that hits the origin.

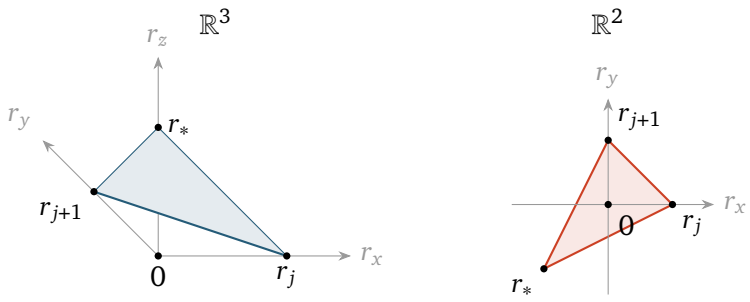


Figure 5.7: For two particles, project to relative coordinates $r = x_1 - x_2$. The spatial triangle misses 0, whereas the planar comparison contains 0.

Coordinate permutations are homeomorphisms commuting with the projection, so they are deck transformations. They act transitively on a fibre and exhaust the deck group by uniqueness from one point. By Corollary 4.6, a simply connected cover whose deck group acts transitively on a fibre has that deck group as the fundamental group of the base. Hence the unordered space has group S_n. The permutation reaching the lifted endpoint is $\rho([\gamma])$, by our chosen action, so this is the isomorphism stated above. Simon, *Topological Quantum*, §3.3.2, describes the exchange-space conclusion.

For the plane we sketch the identification. Lifting a configuration loop labels its strands. Conversely, their unordered positions give a configuration loop. The same correspondence applies to homotopies. Thus the two groups are identified. This is Simon, §3.3.1, in the course's path convention. In particular, the two-particle calculation gives $B_2 \cong \mathbb{Z}$.

5.6 Braid relations and exchange phases

Arrange the base positions on a real line. Let σ_i exchange positions i and $i+1$ counter-clockwise, fixing the other positions. Its inverse exchanges them clockwise. We cite Artin's presentation:

$$B_n = \left\langle \sigma_1, \dots, \sigma_{n-1} \;\middle|\; \begin{array}{ll} \sigma_i\sigma_j = \sigma_j\sigma_i, & |i-j| \geq 2, \\ \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, & 1 \leq i \leq n-2 \end{array} \right\rangle .$$

The first relation exchanges separated pairs in either order. The second changes the order of three adjacent exchanges. The completeness of these relations is cited from Simon, §3.3.1 and Exercise 3.1. Our words are read leftmost first and positive exchanges are counter-clockwise.[1] The elementary crossings below fix the projection convention.

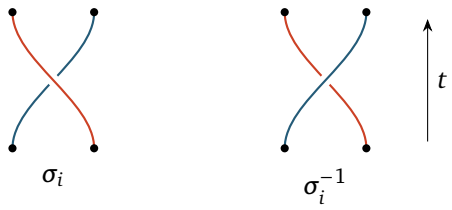


Figure 5.8: Time runs upward and the real axis to the right. At a crossing, the strand with larger imaginary coordinate is drawn on top.

The next figure shows both braid relations with the same labelled strands. At a crossing in the drawing, the particles have different imaginary coordinates, so they do not collide.

[1] Simon reads words rightmost first and takes clockwise exchanges as positive. His positive generator is our σ_i^{-1}. A homomorphism to the group of unit complex numbers with value $e^{i\theta}$ on our positive exchange has value $e^{-i\theta}$ on his positive exchange.

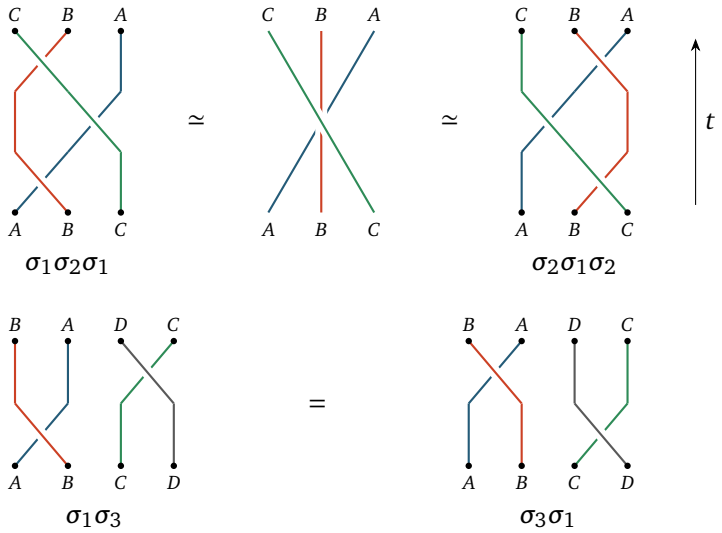


Figure 5.9: Time increases upward, so each word starts at the bottom. The leftward-moving strand is on top at each crossing.

For the three-strand relation, each pair crosses once: C over B, C over A and B over A. Moving the middle crossing past the third strand deforms one diagram into the other with endpoints fixed. The map $\rho\colon B_n \to S_n$ sends σ_i to $(i\ i+1)$. It is surjective because adjacent transpositions generate S_n. It forgets the motion and retains the endpoint permutation. Two positive exchanges already give a nontrivial element of B_2 with identity permutation, by the winding calculation above.

Physical input. We consider a scalar statistical-transport sector for identical particles. The statistical factor multiplies a quantum amplitude, a complex number whose squared modulus gives a probability in the quantum description. Fix a base configuration $x_0 \in \mathrm{Conf}_n(\mathbb{R}^d)$. We assume that each collision-free loop based at x_0 has a unit statistical factor. We assume that this factor depends only on endpoint-fixed homotopy, that successive loops multiply their factors, and that the constant loop has factor 1. This specifies the statistical part of transport after any phase depending on the detailed dynamics has been separated.

These assumptions define a homomorphism from the configuration-space fundamental group to the unit complex numbers. They do not follow from the return of the particle positions. A particular physical system must supply such a sector before the classification applies. For scalar factors, reversing composition order leaves the product unchanged.

Definition 5.19 (Unitary character and exchange phase)**.** A one-dimensional unitary representation is a homomorphism $\chi\colon G \to U(1)$, where $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. Such a representation is also called a *unitary character*. In the scalar statistical sector, its value on an exchange is the *exchange phase*.

Corollary 5.20 (Scalar exchange phases · *proved given the presentations*)**.** *For $n \geq 2$, the one-dimensional unitary representations of S_n give exchange phases $+1$ or -1. For B_n, any phase $e^{i\theta}$, with $\theta \in \mathbb{R}$, is possible, with the same phase on every σ_i.*

Proof. Adjacent transpositions generate S_n and are conjugate to one another. Scalar images of conjugate elements agree. Each transposition squares to the identity, so its common image q satisfies $q^2 = 1$. Both possibilities occur: the constant character and the sign character, the latter given by the determinant of the permutation matrix.

For B_n, write $q_i = \chi(\sigma_i)$. All q_i are nonzero scalars. The adjacent braid relation gives

$$q_i^2 q_{i+1} = q_i q_{i+1}^2, \qquad q_i = q_{i+1}.$$

For $n \geq 3$ this makes all images equal. For $n = 2$ there is only one generator. Conversely, assigning the same $q \in U(1)$ to every generator satisfies both families of defining relations. Thus every $q = e^{i\theta}$ defines a character. ∎

The choices $q = 1$ and $q = -1$ describe scalar bosonic and fermionic statistics. The other planar choices describe *abelian anyons*. Higher-dimensional unitary representations of S_n give matrix-valued exchange operators and the possibility of parastatistics. The scalar conclusion does not classify those representations.

Self-check. State in one sentence what scalar statistical transport is and why it was introduced. For a planar sector with $\chi(\sigma_i) = e^{\pi i/3}$, compute the factor for $\beta = \sigma_1^2\sigma_2^{-1}\sigma_1\sigma_2^{-3}$ in B_3. Determine whether this sector is possible in three dimensions.

Answer. Scalar statistical transport assigns deformation-invariant multiplicative unit factors to collision-free loops, so that topology can classify the statistical factors multiplying quantum amplitudes. The exponent sum of β is $2 - 1 + 1 - 3 = -1$, so

$$\chi(\beta) = e^{-\pi i/3}.$$

In space a double exchange is trivial. The proposed factor fails this relation because $(e^{\pi i/3})^2 = e^{2\pi i/3} \neq 1$. Thus this scalar sector does not descend to three-dimensional exchanges.

Summary. Van Kampen combines path-connected open pieces with path-connected intersection by imposing the overlap relations. A polygonal surface has one relation, the product of its handle commutators. Planar configuration loops are braids. In three dimensions the ordered-space contraction reduces loop classes to permutations. The inverse outgoing permutation is the homomorphism in our path convention. Under the scalar statistical-transport assumptions, exchange phases are arbitrary for braids and restricted to ± 1 for permutations. Lecture 6 counts holes of every dimension by linear algebra.

Reading. Hatcher, *Algebraic Topology*, §1.2; Nakahara, *Geometry, Topology and Physics*, §4.4; Simon, *Topological Quantum*, §§3.3–3.5 and Exercise 3.1.